

SELF-REFINE GUARDRAIL: A TWO-GATE POLICY, CONFIDENTIALITY, AND ERROR VERIFICATION LAYER FOR GENERATIVE AI CUSTOMER SUPPORT CHATBOTS

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ABSTRACT

The increasing deployment of Generative AI in customer-support systems introduces risks involving policy violations, factual inconsistencies, inappropriate communication, and confidentiality exposure. This paper presents **Self-Refine Guardrail (SRG)**, a multi-stage automated review layer designed to evaluate AI-generated responses before customer delivery. SRG employs a Review Agent for policy compliance, communication tone, and response completeness, followed by an Error-Checking Assistant responsible for factual consistency and sensitive-information detection. A deterministic Boolean routing mechanism approves responses only when both review stages satisfy their respective conditions.

The study specifically investigates whether architectural separation between review stages guarantees **reasoning-level scope separation**. An initial eight-scenario evaluation achieved correct routing for all tested cases; however, overlapping policy information revealed that the Review Agent could reason about concerns assigned to the downstream stage. A five-round diagnostic progressively applied exclusion instructions, stronger rules, Boolean verdicts, visible redaction, and seamless redaction. Under the final configuration, all 10 trials were correctly blocked, while confidentiality-flavored reasoning still appeared in 9 of 10 runs. These findings indicate that automated AI review is feasible, but formal architectural boundaries do not necessarily constrain model reasoning. The study therefore identifies reasoning-level scope leakage as an important limitation requiring broader evaluation before production deployment. The results also motivate systematic testing across diverse scenarios, adversarial prompts, multi-turn conversations, and multiple language models to determine the robustness and generalizability of the observed behavior. Future evaluation will focus on quantitative precision, recall, F1, false-positive and false-negative rates, together with confidence intervals, to establish statistically stronger evidence for practical deployment.

Keywords: *Large Language Models, AI Guardrails, Self-Refinement, Policy Compliance, Confidential Data Detection, Retrieval-Augmented Generation, LLM Safety, Customer Support Automation*